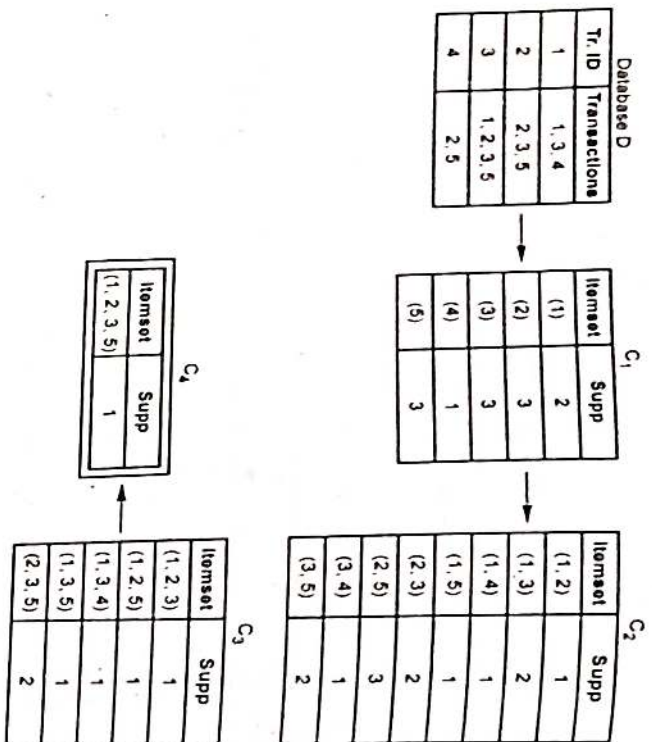


Ans. :



END...

Unit VI

6

Introduction to Artificial Neural Network

6.1 : Perceptron Learning : Biological Neuron

Q.1 What is perceptron ?

Ans. : • An arrangement of one input layer of McCulloch-Pitts neurons feeding forward to one output layer of McCulloch-Pitts neurons is known as a perceptron.

• The perceptron is a feed - forward network with one output neuron that learns a separating hyper - plane in a pattern space. Fig. Q.1.1 shows perceptron.

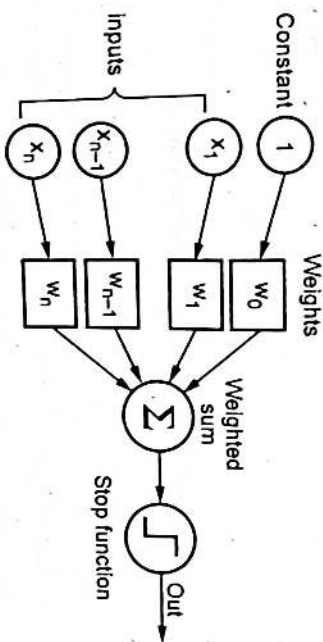


Fig. Q.1.1 Perceptron

- Perceptron is an Artificial Neuron. It is the simplest possible Neural Network. The original Perceptron was designed to take a number of binary inputs, and produce one binary output (0 or 1).
- The idea was to use different weights to represent the importance of each input, and that the sum of the values should be greater than a threshold value before making a decision like true or false (0 or 1).

Q.2 What is biological neuron ? Explain with diagram and its components.

Ans : • Biological neurons, consisting of a cell body, axons, dendrites and synapses, are able to process and transmit neural activation.

• Artificial neural systems are inspired by biological neural systems. The elementary building block of biological neural systems is the neuron.

• The brain is a collection of about 10 billion interconnected neurons. Each neuron is a cell [right] that uses biochemical reactions to receive, process and transmit information. Fig. Q.2.1 shows biological neural systems.

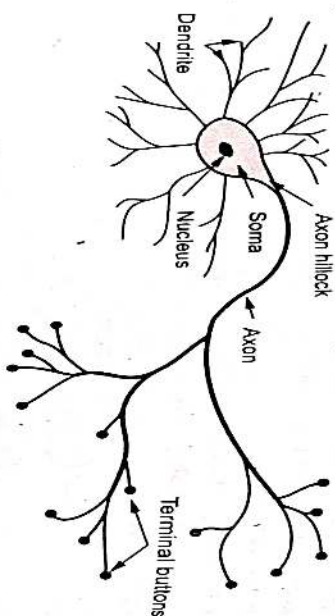


Fig. Q.2.1 Schematic of biological neuron

• The single cell neuron consists of the cell body or soma, the dendrites and the axon. The dendrites receive signals from the axons of other neurons. The small space between the axon of one neuron and the dendrite of another is the synapse. The afferent dendrites conduct impulses toward the soma. The efferent axon conducts impulses away from the soma.

Basic Components of Biological Neurons

1. The majority of *neurons* encode their activations or outputs as a series of brief electrical pulses (i.e. spikes or action potentials).
2. The neuron's *cell body (soma)* processes the incoming activations and converts them into output activations.
3. The neuron's *nucleus* contains the genetic material in the form of DNA. This exists in most types of cells, not just neurons.

4. **Dendrites** are fibres which emanate from the cell body and provide the receptive zones that receive activation from other neurons.

5. **Axons** are fibres acting as transmission lines that send activation to other neurons.

6. The junctions that allow signal transmission between the axons and dendrites are called **synapses**. The process of transmission is by diffusion of chemicals called **neurotransmitters** across the synaptic cleft.

Q.3 Compare biological NN with artificial NN.

Ans : Comparison between Biological NN and Artificial NN

Biological NN	Artificial NN
Soma	Unit
Axon, dendrite	Dendrite
Synapse	Weight
Potential	Weighted Sum
Threshold	Bias Weight
Signal	Activation

6.2 : Introduction to ANN

Q.4 What is artificial neural network ? List its characteristics.

Ans : • Artificial Neural Network (ANN) is a computational system inspired by the structure, processing method, learning ability of a biological brain. An artificial neural network is composed of many artificial neurons that are linked together according to specific network architecture. The objective of the neural network is to transform the inputs into meaningful outputs.

• ANNs do not execute programmed instructions; they respond in parallel to the pattern of inputs presented to it. There are also no separate memory addresses for storing data. Instead, information is contained in the overall activation 'state' of the network. 'Knowledge' is thus

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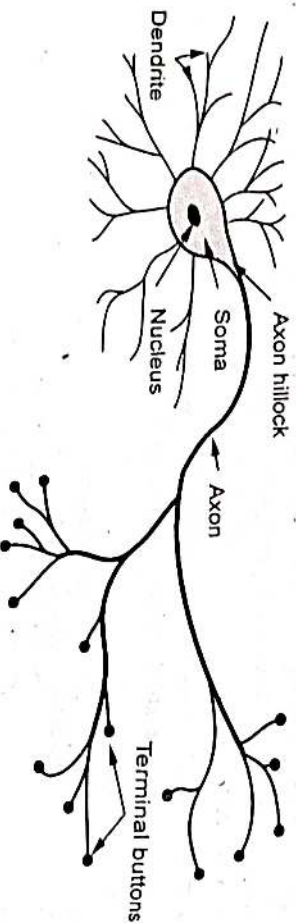


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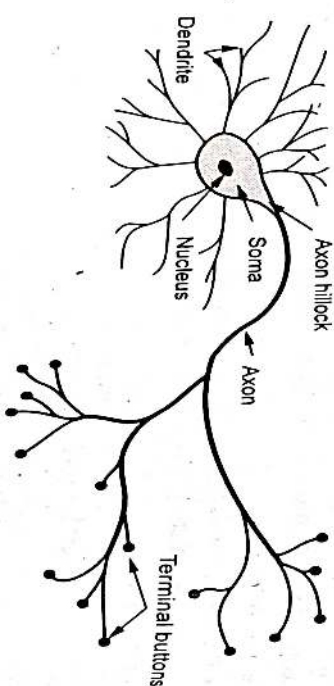


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• ANNs do not execute programmed instructions; they respond in parallel to the pattern of inputs presented to it. There are also no separate memory addresses for storing data. Instead, information is contained in the overall activation 'state' of the network. 'Knowledge' is thus

represented by the network itself, which is quite literally more than the sum of its individual components.

- Fig Q.4.1 shows artificial neural network.

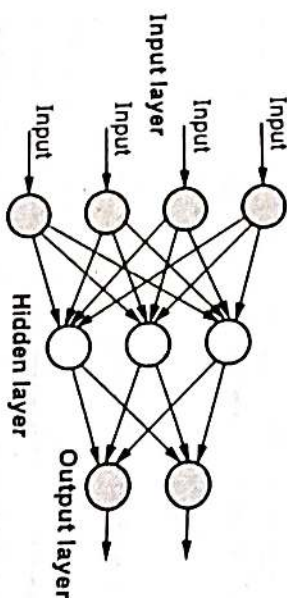


Fig. Q.4.1 Artificial neural network

- Elements of ANN are processing units, topology and learning algorithm.
- Tasks to be solved by artificial neural networks :
 1. Controlling the movements of a robot based on self-perception and other information;
 2. Deciding the category of potential food items in an artificial world;
 3. Recognizing a visual object;
 4. Predicting where a moving object goes, when a robot wants to catch it.

- Characteristics of artificial neural networks

1. Large number of very simple processing neuron-like processing elements.
2. Large number of weighted connections between the elements.
3. Distributed representation of knowledge over the connections.
4. Knowledge is acquired by network through a learning process.

Q.5 Explain advantages and application of neural network.

Ans. : Advantages of neural network

The advantages of neural networks are due to its adaptive and generalization ability.

- a) Neural networks are adaptive methods that can learn without any prior assumption of the underlying data.
- b) Neural network, namely the feed forward multilayer perception and radial basis function network have been proven to be universal functional approximations.
- c) Neural networks are non-linear model with good generalization ability.

Applications of Neural Network

Neural network applications can be grouped in following categories :

1. **Clustering** : A clustering algorithm explores the similarity between patterns and places similar patterns in a cluster. Best known applications include data compression and data mining.
2. **Classification/Pattern recognition** : The task of pattern recognition is to assign an input pattern (like handwritten symbol) to one of many classes. This category includes algorithmic implementations such as associative memory.
3. **Function approximation** : The tasks of function approximation is to find an estimate of the unknown function $f()$ subject to noise. Various engineering and scientific disciplines require function approximation.
4. **Prediction/Dynamical systems** : The task is to forecast some future values of a time-sequenced data. Prediction has a significant impact on decision support systems. Prediction differs from function approximation by considering time factor.

Q.6 What are the characteristics of ANN ?

Ans. : Characteristics of Artificial Neural Networks :

1. Large number of very simple processing neuron-like processing elements.
2. Large number of weighted connections between the elements.
3. Distributed representation of knowledge over the connections.
4. Knowledge is acquired by network through a learning process.

Q.7 Explain with example, the challenges in assigning synaptic weights for the interconnection between neurons ? How can this challenge be addressed ?

Ans. : • Artificial neural networks are specifically designed for a particular function like binary classification, multi class classification,

pattern recognition and so on through learning processes. The weights of the synaptic connections of both the neural networks adjust with the learning process.

- Each connection between two neurons has a unique synapse with a unique weight attached to it. When we talk about updating weights in a network, we're really talking about adjusting the weights on these synapses.

- Weights are values that control the strength of the connection between two neurons. That is, inputs are typically multiplied by weights, and that defines how much influence the input will have on the output.

- In other words, when the inputs are transmitted between neurons, the weights are applied to the inputs along with an additional value (the bias).

- The connection is controlled by the strength or amplitude of a connection between both nodes, also called the synaptic weight. Multiple synapses can connect the same neurons, with each synapse having a different level of influence (trigger) on whether that neuron is "fired" and activates the next neuron.

- the synapse is represented as a weight vector. The weights control how the output of the previous layer is fed through the activation function for each node. Using the convenient scalability of linear algebra, all the weight vectors of the synapses for an entire node layer can be represented as a single weight matrix.

- The resulting output of all these nodes is used as the input for the next layer of nodes and so on throughout the neural network "brain" until the final output layer is reached.

- The synapse is defined by these weights and expressed algebraically (for a linear synapse) as :

$$Y_i = \sum_j w_{ij} X_j \quad \text{or} \quad Y = WX$$

- Typically a bias term will be added and the output values will be passed through a choice of activation function before being used as inputs into the subsequent layer

6.3 : McCulloch Pitts Neuron

Q.8 Explain McCulloch Pitts neuron with diagram.

Ans. : • The first mathematical model of a biological neuron was presented by McCulloch and Pitts. This model is known as McCulloch Pitt model. It is basic building block of neural network.

- Directed weight graph is used for connecting neurons.

- McCulloch and Pitts describe a neuron as a logical threshold element with two possible states. Such a threshold element has "N" input channels and one output channel. An input channel is either active (input 1) or silent (input 0).

- The activity states of all input channels thus encode the input information as a binary sequence of N bits. The state of the threshold element is then given by linear summation of all a different input signals x_i and comparison of the sum with a threshold value s .

- The system of neurons is static and acts synchronously. A processor (system) with multiple inputs and a single output.

- Effective input : Weighted sum of all inputs.

- Bias or threshold : If the effective input is larger than the bias, the neuron outputs a one, otherwise, it outputs a zero.

- Fig. Q.8.1 shows McCulloch Pitt model.

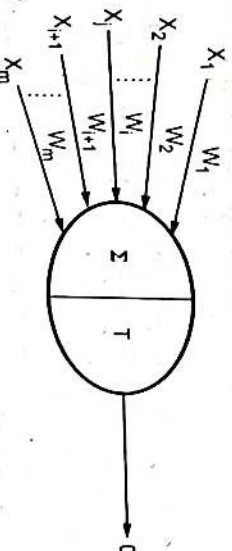


Fig. Q.8.1

- This model can be described in a mathematical formalism as follows :
 $0 = \theta(a)$

Where, $a = \sum W_j X_j - T$
And

$\theta(x)$ is a function such that $\theta(x) = 1$ if $x > 0$, otherwise $\theta(x) = 0$.

- The parameters used to scale the inputs are called the weights. The effective input is the weighted sum of the inputs. The parameter to measure the switching level is the threshold or bias. Neuron fires (output of one) when its net input excitation exceeds a certain value called 'threshold'. Threshold is the minimum value of the sum of the weighted active inputs needed for the postsynaptic neuron to fire.
- The function for producing the final output is called the activation function, which is the step function in the McCulloch-Pitts model.

$$0 = f \left(\sum_{j=1}^N W_j X_j - T \right)$$

$$f(u) = \begin{cases} 1 & \text{if } u \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Their "neurons" operated under the following assumptions :

1. They are binary devices (0,1).
2. Each neuron has a fixed threshold (theta).
3. The neuron receives inputs from excitatory synapses, all having identical weights.
4. Inhibitory inputs have an absolute veto power over any excitatory inputs.
5. At each time step the neurons are simultaneously (synchronously) updated by summing the weighted excitatory inputs and setting the output to 1 iff the sum is greater than or equal to the threshold AND if the neuron receives no inhibitory input.

6.4 : Perceptron and Its Learning Algorithm

Q.9 Briefly discuss Adaline network.

Ans.: ADALINE (Adaptive Linear Neuron) is an early single-layer artificial neural network.

BCODB

- An important generalized of the perceptrons training algorithm was presented by Widrow and Hoff as the least mean square learning procedure also known as the delta rule.
- The learning rule was applied to the "adaptive linear element" also named Adaline.
- The perceptron learning rule uses the output of the threshold function for learning. The delta rule uses the net output without further mapping into output values -1 or +1.

• Fig. Q.9.1 shows adaline.

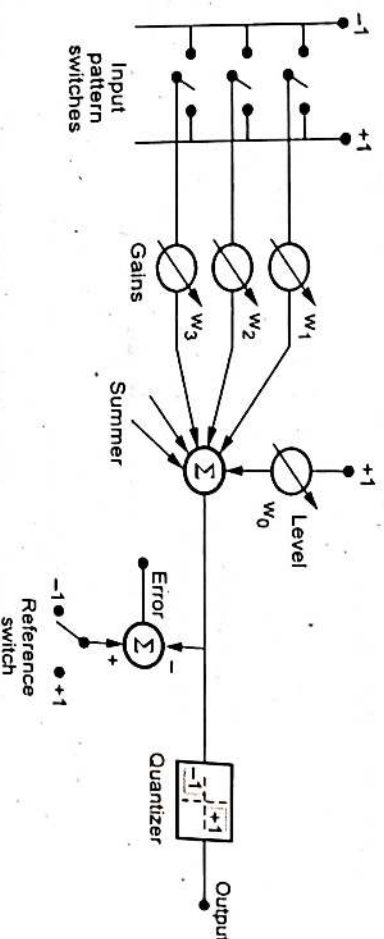


Fig. Q.9.1 Adaline

- If the input conductances are denoted by w_i where $m \ i = 0, 1, 2, \dots, n$ and input and output signals by x_i and y respectively, then the output of the central block is defined to be :

$$y = \sum_{i=1}^n w_i x_i + \theta$$

Where, $\theta \equiv w_0$

- In a simple physical implementation, this device consists of a set of controllable resistors connected to a circuit which can sum up currents caused by the input voltage signals. Usually the central block, the summer is also followed by a quantizer which outputs +1 of -1, depending on the polarity of the sum.

BCODB

- The problem is to determine the coefficients w_i , where $i = 0, 1, \dots, n$, in such way that the input output response is correct for a large number of arbitrarily chosen signal sets.
- If an exact mapping is not possible the average error must be minimized, for instance, in the sense of least squares.
- An adaptive operation means that there exists a mechanism by which the w_i can be adjusted, usually iteratively to attain the correct values.
- For the Adaline, Widrow introduced the delta rule to adjust the weights.
- For the p^{th} input-output pattern, the error measure of a single-output Adaline can be expressed as,

$$E_p = (t_p - o_p)^2$$

Where

t_p = Target output

o_p = Actual output of the Adaline

- The derivation of E_p with respect to each weight w_i is

$$\frac{\partial E_p}{\partial w_i} = -2(t_p - o_p) x_i$$

- To decrease E_p by gradient descent, the update formula for w_i on the p^{th} input-output pattern is

$$\Delta_p w_i = \eta(t_p - o_p) x_i$$

- The delta rule tries to minimize squared errors, it is also referred to as the least mean square learning procedure or Widrow - Hoff learning rule.

Q.10 What is activation function ? Also explain logistic function and Arc tangent.

- Ans. :**
- Activation functions also known as transfer function is used to map input nodes to output nodes in certain fashion.
 - The activation function is the most important factor in a neural network which decided whether or not a neuron will be activated or not and transferred to the next layer.

- Activation functions help in normalizing the output between 0 to 1 or -1 to 1. It helps in the process of backpropagation due to their differentiable property. During backpropagation, loss function gets updated, and activation function helps the gradient descent curves to achieve their local minima.
- Activation function basically decides in any neural network that given input or receiving information is relevant or it is irrelevant.
- These activation function makes the multilayer network to have greater representational power than single layer network only when non-linearity is introduced.
- The input to the activation function is sum which is defined by the following equation.

$$\text{sum} = I_1 W_1 + I_2 W_2 + \dots + I_n W_n = \sum_{j=1}^n I_j W_j + b$$

- **Activation Function : Logistic Function**

$$f(\text{sum}) = \frac{1}{(1 + e^{-s * \text{sum}})} = (1 + e^{-s * \text{sum}})^{-1}$$

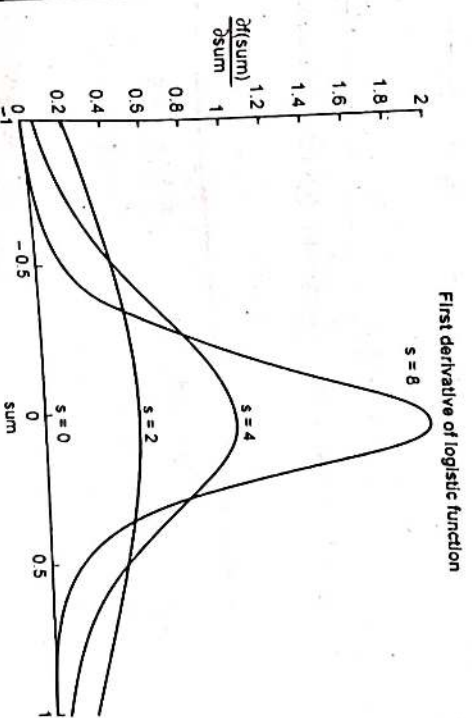
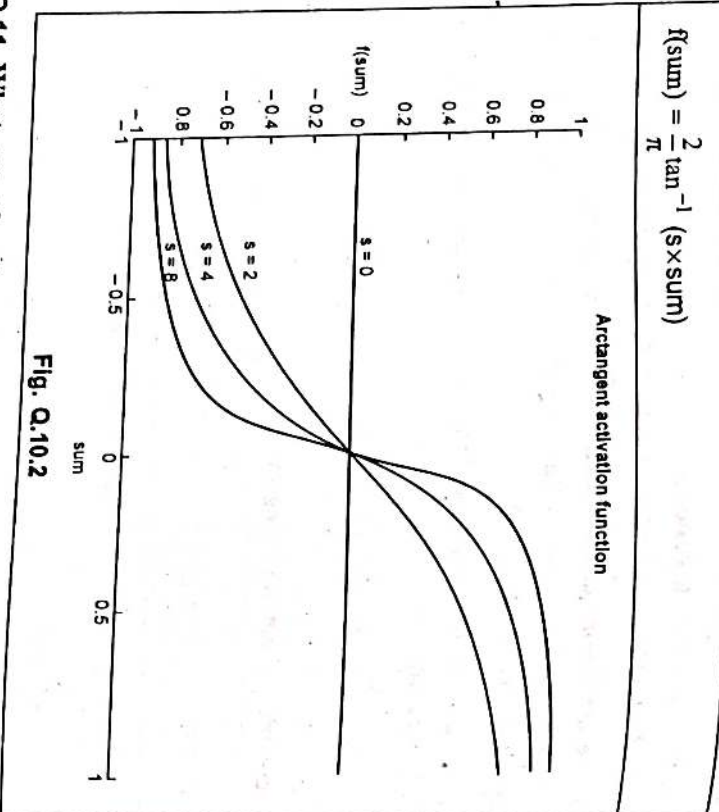


Fig. Q.10.1

- Logistic function monotonically increases from a lower limit (0 or -1) to an upper limit (+1) as sum increases. In which values vary between 0 and 1, with a value of 0.5 when I is zero.

• Activation Function : Arc Tangent

$$f(\text{sum}) = \frac{2}{\pi} \tan^{-1}(s \times \text{sum})$$



Q.11 What are advantages and disadvantages of Sigmoid, Tanh and ReLU.

Ans. :

Function	Advantages	Disadvantages
Sigmoid	1. Output in range (0,1)	1. Saturated Neurons 2. Not zero entered 3. Small gradient 4. Vanishing gradient

Tanh	1. Zero centered, 2. Output in range (-1,1)	1. Saturated Neurons
ReLU	1. Computational efficiency, 2. Accelerated convergence	1. Dead Neurons, 2. Not zero centered

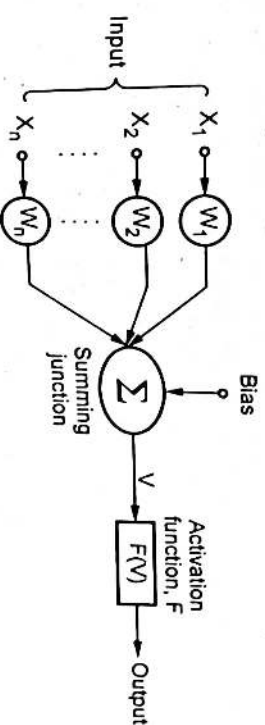
Q.12 What is the function of a summation junction of a neuron? What is threshold activation function?

Ans. : • Summation junction for the input signals is weighted by the respective synaptic weight. Because it is a linear combiner or adder of the weighted input signals, the output of the summation junction (y) can be expressed as follows:

$$y_{\text{sum}} = \sum_{i=1}^n w_i x_i$$

- This summed function is applied over an Activation function. The output from this neuron is multiplied with the weight W_3 and supplied as input to the output layer.

• Fig. Q.12.1 shows summing function and activation function.



- Threshold activation function also known as activation function, it results in an output signal only when an input signal exceeding a specific threshold value comes as an input. It is similar in behaviour to the biological neuron which transmits the signal only when the total input signal meets the firing threshold.

- Activation functions help in normalizing the output between 0 to 1 or -1 to 1. It helps in the process of backpropagation due to their differentiable property. During backpropagation, loss function gets updated, and activation function helps the gradient descent curves to achieve their local minima.
- Activation function basically decides in any neural network that given input or receiving information is relevant or it is irrelevant.
- These activation function makes the multilayer network to have greater representational power than single layer network only when non-linearity is introduced.

6.5 : Multi-Layer Perceptron Model

Q.13 Briefly discuss multilayer perceptron NN.

Ans. : • Multilayer perceptron is a neural network that learns the relationship between linear and non-linear data. Multilayer Perceptron has input and output layers, and one or more hidden layers with many neurons stacked together.

• Multilayer perceptron falls under the category of feed-forward algorithms, because inputs are combined with the initial weights in a weighted sum and subjected to the activation function.

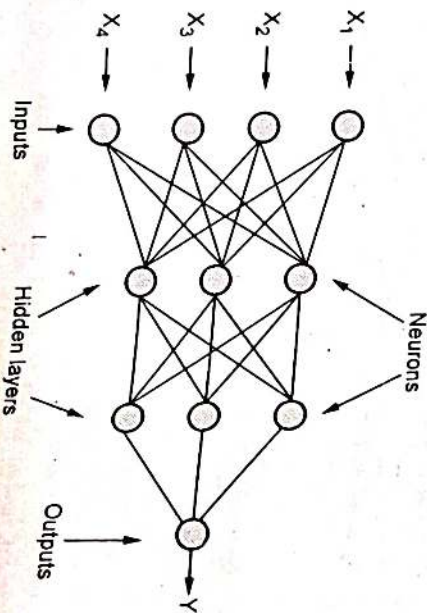


Fig. Q.13.1 Multilayer perceptron NN

- Fig. Q.13.1 shows Multilayer perceptron NN.
- Each layer is feeding the next one with the result of their computation, their internal representation of the data. This goes all the way through the hidden layers to the output layer.

• Multilayer perceptron is a typical example of a feedforward artificial neural network.

• The number of layers and the number of neurons are referred to as hyper-parameters of a neural network, and these need tuning. Cross-validation techniques must be used to find ideal values for these.

• The weight adjustment training is done via back-propagation. Deeper neural networks are better at processing data. However, deeper layers can lead to vanishing gradient problems. Special algorithms are required to solve this issue.

• The Multilayer Perceptron learning procedure is as follows:

- Starting with the input layer, propagate data forward to the output layer. This step is the forward propagation.
- Based on the output, calculate the error. The error needs to be minimized.
- Backpropagate the error. Find its derivative with respect to each weight in the network, and update the model.
- Repeat the three steps given above over multiple epochs to learn ideal weights.
- Finally, the output is taken via a threshold function to obtain the predicted class labels.

Q.14 Explain back-propagation neural networks.

Ans. : • Backpropagation is a training method used for a multi-layer neural network. It is also called the generalized delta rule. It is a gradient descent method which minimizes the total squared error of the output computed by the net.

- The backpropagation algorithm looks for the minimum value of the error function in weight space using a technique called the delta rule or

gradient descent. The weights that minimize the error function is then considered to be a solution to the learning problem.

- Backpropagation is a systematic method for training multiple layer ANN. It is a generalization of Widrow-Hoff error correction rule. 80 % of ANN applications uses backpropagation.

- Fig. Q.14.1 shows backpropagation network.

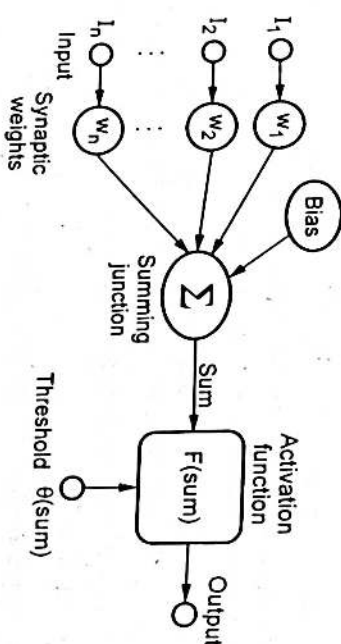


Fig. Q.14.1 Backpropagation network

- Consider a simple neuron :

- Neuron has a summing junction and activation function.
- Any non linear function which differentiable everywhere and increases everywhere with sum can be used as activation function.
- Examples : Logistic function, Arc tangent function, Hyperbolic tangent activation function.

- These activation function makes the multilayer network to have greater representational power than single layer network only when non-linearity is introduced.

- Weights connects unit (neuron) in one layer only to those in the next higher layer. The output of the unit is scaled by the value of the connecting weight and it is fed forward to provide a portion of the activation for the units in the next higher layer.

- Backpropagation can be applied to an artificial neural network with any number of hidden layers. The training objective is to adjust the weights so that the application of a set of inputs produces the desired outputs.

RECORD

Q.15 Which factors are influencing backpropagation training ?

Ans. : Factors Influencing backpropagation training :

- The training time can be reduced by using :

 - Bias :** Networks with biases can represent relationships between inputs and outputs more easily than networks without biases. Adding a bias to each neuron is usually desirable to offset the origin of the activation function. The weight of the bias is trainable similar to weight except that the input is always +1.

- Momentum :** The use of momentum enhances the stability of the training process. Momentum is used to keep the training process going in the same general. In backpropagation with momentum, the weight change is a combination of the current gradient and the previous gradient.

Q.16 Explain advantages and disadvantages of backpropagation.

Ans. : Advantages of backpropagation

- It is simple, fast and easy to program.
- Only numbers of the input are tuned and not any other parameter.
- No need to have prior knowledge about the network.
- It is flexible.
- A standard approach and works efficiently.
- It does not require the user to learn special functions.

Disadvantages of backpropagation

- Backpropagation possibly be sensitive to noisy data and irregularity.
- The performance of this is highly reliant on the input data.
- Needs excessive time for training.
- The need for a matrix-based method for backpropagation instead of mini-batch.

Q.17 What is need of hidden layer in backpropagation ? How hidden units are selected ?

Ans. : • Need of hidden layers

- A network with only two layers (input and output) can only represent the input with whatever representation already exists in the input data.

RECORD

2. If the data is discontinuous or non-linearly separable, the innate representation is inconsistent and the mapping cannot be learned using two layers (Input and Output).
3. Therefore, hidden layer(s) are used between input and output layers.
- **Selection of number of hidden units :** The number of hidden units depends on the number of input units.
 1. Never choose h to be more than twice the number of input units.
 2. You can load p patterns of 1 elements into $\log_2 p$ hidden units.
 3. Ensure that we must have at least $1/e$ times as many training examples.
 4. Features extraction requires fewer hidden units than inputs.
 5. Learning many examples of disjointed inputs requires more hidden units than inputs.
 6. The number of hidden units required for a classification task increases with the number of classes in the task. Large network require longer training times.

Q.18 Write short note on recurrent neural network.

Ans. : • A recurrent neural network is a type of neural network that contains loops, allowing information to be stored within the network.

- A RNN is particularly useful when a sequence of data is being processed to make a classification decision or regression estimate but it can also be used on non-sequential data. Recurrent neural networks are typically used to solve tasks related to time series data.
- Applications of recurrent neural networks include natural language processing, speech recognition, machine translation, character-level language modeling, image classification, image captioning, stock prediction, and financial engineering.
- Fig. Q.18.1 shows architecture of recurrent neural network.
- Recurrent Neural Networks can be thought of as a series of networks linked together. They often have a chain-like architecture, making them applicable for tasks such as speech recognition, language translation, etc.

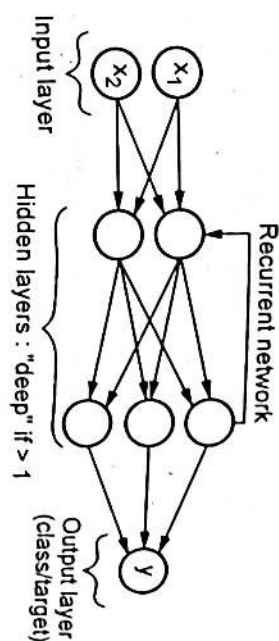


Fig. Q.18.1

- An RNN can be designed to operate across sequences of vectors in the input, output, or both. For example, a sequenced input may take a sentence as an input and output a positive or negative sentiment value. Alternatively, a sequenced output may take an image as an input, and produce a sentence as an output.

6.6 : Learning Parameter : Bias and Weight

Q.19 Discuss learning parameter weight and bias and compare bias with weight.

Ans. : • Weights and biases are the learnable parameters that help a neural network correctly learn a function. A machine learning model uses lots of examples to learn the correct weights and bias to assign to each feature in a dataset to help it correctly predict outputs.

- Weight is the parameter within a neural network that transforms input data within the network's hidden layers. A neural network is a series of nodes, or neurons. Within each node is a set of inputs, weight, and a bias value.
- As an input enters the node, it gets multiplied by a weight value and the resulting output is either observed, or passed to the next layer in the neural network. Often the weights of a neural network are contained within the hidden layers of the network.
- Choose the weights w_i to best fit the set of training examples.
- Minimize the squared error E between the train values and the values predicted by the hypothesis

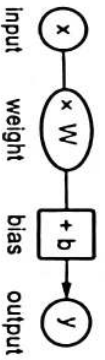
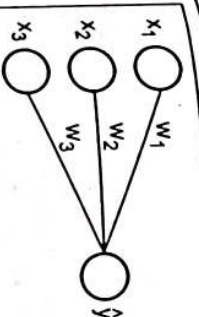


Fig. Q.19.1

$$E = \sum (V_{\text{train}}(b) - \hat{Y}(b))^2$$

(b, $V_{\text{train}}(b) \in \text{training examples}$)

- Require an algorithm that will incrementally refine weights as new training values.
- Least Mean Squares (LMS) is one such algorithm.
- Weight vs. Bias**
- A teachable neural network will randomize both the weight and bias values before learning initially begins. As training continues, both parameters are adjusted toward the desired values and the correct output.
- The two parameters differ in the extent of their influence upon the input data. Simply, bias represents how far off the predictions are from their intended value.
- Biases make up the difference between the function's output and its intended output. A low bias suggest that the network is making more assumptions about the form of the output, whereas a high bias value makes less assumptions about the form of the output.
- Weights, on the other hand, can be thought of as the strength of the connection. Weight affects the amount of influence a change in the input will have upon the output. A low weight value will have no significantly change the output.
- In most learning networks, error is calculated as the difference between the actual output y and the predicted output $\hat{y}$. Loss functions are quantitative measures of how satisfactory the model predictions are.



Output :

$$\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b = w^T x + b$$

Neural network parameters

$$W = \{w_1, w_2, w_3\} b$$

The function that is used to compute this error is known as Loss Function also known as Cost function.

6.7 : Introduction to Deep Learning

Q.20 Explain multi task learning.

[SPPU : May-16, 17, (End Sem), Marks 8]

OR Explain multi-task learning with task grouping and overlap methodology.

[SPPU : May-18, (End Sem), Marks 8]

Ans. : • Multi-task learning refers to learning multiple tasks simultaneously, in order to avoid tabula rasa learning and to share information between similar tasks during learning.

- The transfer of information between tasks is usually eased by learning the tasks with similar machine learning models or algorithms under common input and output representations. The aim may be to increase the performance on all tasks, or it may be to increase the performance on one task of primary interest given other tasks of secondary interests.

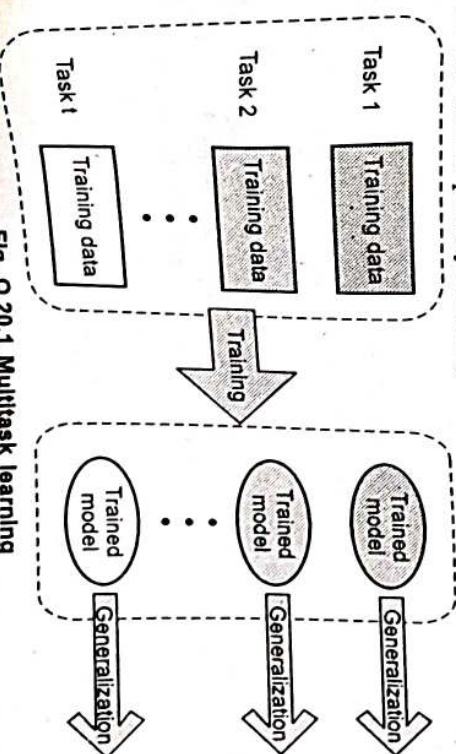


Fig. Q.20.1 Multitask learning

• Fig. Q.20.1 shows multitask learning.

• To differentiate between these two cases of multi-task learning, call them symmetric and asymmetric respectively. In the machine learning literature, it is more common to find symmetric multi-task learning. In contrast, asymmetric multi-task learning is more prevalent in the geo-statistics community, where the secondary tasks pertain to information more readily available than the primary task.

• For example, the primary task may be to predict the concentration of a precious metal, while the secondary tasks provide the concentrations of common metals.

• Example : Web pages categorization

1. Classify documents into categories
2. The classification of each category is a task
3. The tasks of predicting different categories may be latently related.

Q.21 Explain deep learning. What are the challenges in deep learning ?

☞ [SPPU : May-18, (End Sem), Marks 7]

OR Explain deep learning.

☞ [SPPU : May-16, (End Sem), Marks 8]

OR Write a note on deep learning.

☞ [SPPU : Dec-16, (End Sem), Marks 8]

Ans. : • Deep Learning is a new area of machine learning research, which has been introduced with the objective of moving machine learning closer to one of its original goals.

• Deep learning is about learning multiple levels of representation and abstraction that help to make sense of data such as images, sound, and text.

• 'Deep learning' means using a neural network with several layers of nodes between input and output. It is generally better than other methods on image, speech and certain other types of data because the series of layers between input and output do feature identification and processing in a series of stages, just as our brains seem to.

• Deep learning emphasizes the network architecture of today's most successful machine learning approaches. These methods are based on "deep" multi-layer neural networks with many hidden layers.

Challenges In Deep learning :

- They need to find and process massive datasets for training
- One of the reasons deep learning works so well is the large number of interconnected neurons, or free parameters, that allow for capturing subtle nuances and variations in data
- Due to the sheer number of layers, nodes, and connections, it is difficult to understand how deep learning networks arrive at insights
- Deep-learning networks are highly susceptible to the butterfly effect-small variations in the input data can lead to drastically different results, making them inherently unstable

Q.22 Write a note on deep learning and its applications.

☞ [SPPU : Dec-17, (End Sem), Marks 7]

Ans. : Refer Q.21 of Chapter 6.

Application :

1. Colorization of black and white images.
2. Adding sounds to silent movies.
3. Automatic machine translation.
4. Object classification in photographs.
5. Automatic handwriting generation.
6. Character text generation.
7. Image caption generation.
8. Automatic game playing.

END... ✍

SOLVED MODEL QUESTION PAPER (In Sem)

Machine Learning

T.E. (IT) Semester - V [As Per 2019 Pattern]

Time : 1 Hour]

[Maximum Marks : 30

N.B. : i) Attempt Q.1 or Q.2, Q.3 or Q.4.

- ii) Neat diagrams must be drawn wherever necessary.
- iii) Figures to the right side indicate full marks.
- iv) Assume suitable data, if necessary.

Q.1 a) What is machine learning ? Explain types of machine learning. (Refer Q.1 of Chapter - 1) [5]

b) Differentiate between supervised and unsupervised learning. (Refer Q.9 of Chapter - 1) [4]

c) What is the need if dimensionality reduction ? Explain subset selection. (Refer Q.22 of Chapter - 1) [6]
OR

Q.2 a) Explain predictive and descriptive task. (Refer Q.6 of Chapter - 1) [5]

b) Explain qualitative data types. (Refer Q.12 of Chapter - 1) [3]

c) What is feature selection ? What is role of feature selection in ML ? Explain feature selection algorithm. (Refer Q.19 of Chapter - 1) [7]

Q.3 a) Explain any 5 binary classification performance evaluation parameters (excluding accuracy and error rate). (Refer Q.3 of Chapter - 2) [5]

b) Write short note on F-Measure. (Refer Q.12 of Chapter - 2) [4]

(M - 1)

c) Explain kernel methods for non-linearity. (Refer Q.20 of Chapter - 2) [6]
OR

Q.4 a) State formulae for calculating accuracy, true positive rate, true negative rate, false positive rate and false negative rate for binary classification tasks. (Refer Q.4 of Chapter - 2) [7]

b) Explain various multiclass classification techniques. (Refer Q.14 of Chapter - 2) [8]

SOLVED MODEL QUESTION PAPER (End Sem)

Machine Learning

T.E. (IT) Semester - V [As Per 2019 Pattern]

Time : $2 \frac{1}{2}$ Hours]

[Maximum Marks : 70

N.B. : i) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.

- ii) Neat diagrams must be drawn wherever necessary.
- iii) Figures to the right side indicate full marks.
- iv) Assume suitable data, if necessary.

Q.1 a) Consider following data for 5 students.

Each X_i ($i = 1$ to 5) represents the score of i^{th} student in standard X and corresponding Y_i ($i = 1$ to 5) represents the score of i^{th} student in standard XII.

- i) What linear regression equation best predicts standard XIIth score ?
- ii) Find regression line that fits best for given sample data.
- iii) How to interpret regression equation ?
- iv) If a student's score is 80 in std X, then what is his expected score in XII standard ? (Refer Q.6 of Chapter - 3) [3]

Student	Score in X standard (Xi)	Score in Y standard (Yi)
1	95	85
2	85	95
3	80	70
4	70	65
5	60	70

- b) When is it suitable to use linear regression over classification? (Refer Q.3 of Chapter - 3) [8]

- c) Explain gradient descent algorithm. Also explain its limitation. (Refer Q.14 of Chapter - 3) [5]

OR

- Q.2 a) Consider following data :

- i) Find values of β_0 and β_1 w.r.t. linear regression model which best fits given data.

- ii) Interpret and explain equation of regression line.

- iii) If new person rates "Bahubali -Part I" as 3 then predict the rating of same person for "Bahubali -Part II". (Refer Q.7 of Chapter - 3)

Person	X_i = Rating for movie "Bahubali Part - I" by i^{th} person	Y_i = Rating for movie "Bahubali Part - II" by i^{th} person
1 st	4	3
2 nd	2	4
3 rd	3	2
4 th	5	5
5 th	1	3
6 th	3	1

DECODE

[8]

- b) Define and explain Squared Error (SE) and Mean Squared Error (MSE) w.r.t. Regression. (Refer Q.8 of Chapter - 3) [5]

- c) What is a polynomial regression? How it can be represented in a form of a matrix. (Refer Q.21 of Chapter - 3) [5]

- Q.3 a) Write the Grow Tree algorithm to generate feature tree? Explain the role of best split in this algorithm. (Refer Q.7 of Chapter - 4) [10]

- b) Write and explain Naïve Bayes classification algorithm. (Refer Q.13 of Chapter - 4) [7]

OR

- Q.4 a) What is a probabilistic model? Give an example of it. (Refer Q.10 of Chapter - 4) [5]

- b) Write short on regression tree with its algorithm. (Refer Q.3 of Chapter - 4) [5]

- c) What is decision tree? Explain. (Refer Q.2 of Chapter - 4) [7]

- Q.5 a) Explain kNN algorithm with its advantages and disadvantages. (Refer Q.5 of Chapter - 5) [10]

- b) Write a note on clustering trees. (Refer Q.9 of Chapter - 5) [8]

OR

- Q.6 a) Discuss various distance measures. (Refer Q.1 of Chapter - 5) [10]

- b) What is Apriori algorithm? Explain use and limitation of Apriori algorithm. (Refer Q.24 of Chapter - 5) [8]

DECODE

Q.7 a) What is the function of a summation junction of a neuron ?
What is threshold activation function ? (Refer Q.12 of Chapter - 6) [5]

b) Which factors are influencing backpropagation training ?
(Refer Q.15 of Chapter - 6) [4]

c) Explain McCulloch Pitts neuron with diagram.
(Refer Q.8 of Chapter - 6) [8]

OR

Q.8 a) Write short note on recurrent neural network.
(Refer Q.18 of Chapter - 6) [4]

b) Briefly discuss multilayer perceptron NN.
(Refer Q.13 of Chapter - 6) [5]

c) What is artificial neural network ? List its characteristics.
(Refer Q.4 of Chapter - 6) [8]

END... ✍

Notes